Agentic AI Hackathon 2026

SahAI

SAHAYIKA + AI

An offline-first AI co-pilot that helps India's ASHA workers see a maternal health risk coming — before it becomes an emergency.

ANANYA BIJU • ANN SANI • ALVINO P SHATTO

THE PROBLEM STATEMENT

88

Maternal deaths / 100,000 live births, 2021-23

SOURCE: SRS, REGISTRAR GENERAL OF INDIA

86%

Reduction in maternal mortality since 1990

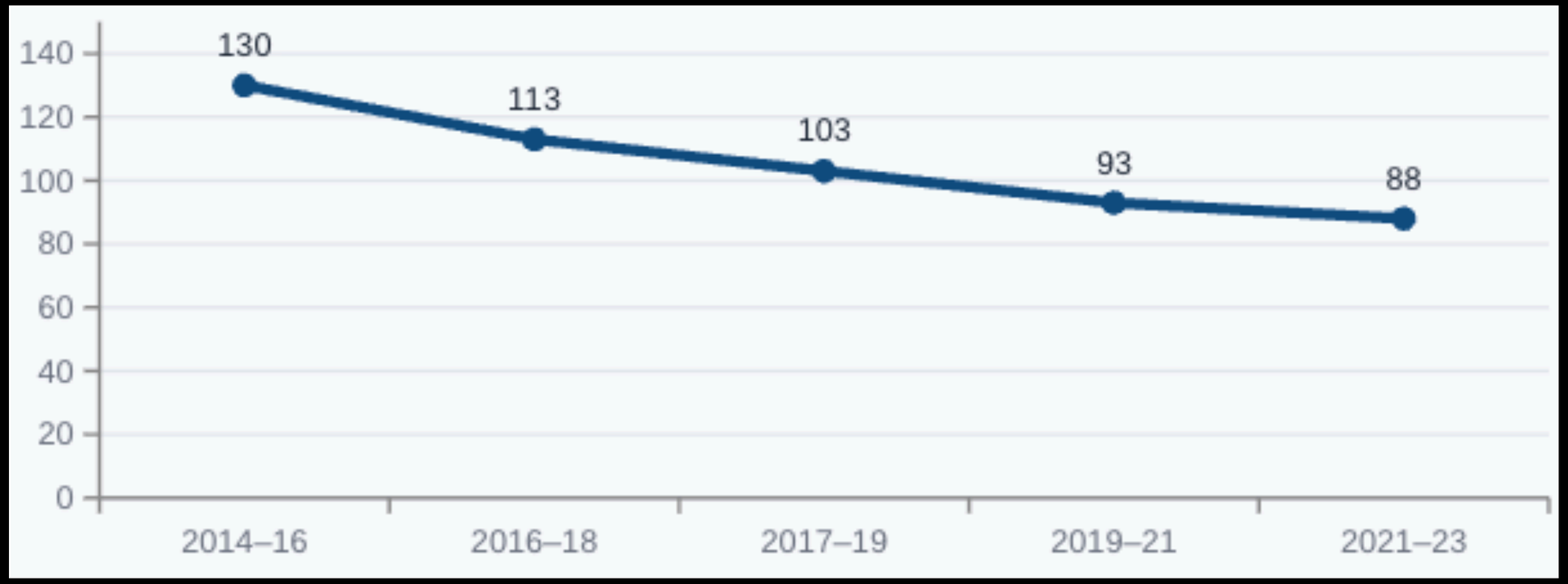
UN MATERNAL MORTALITY ESTIMATION INTER-AGENCY GROUP (2025)

30

MMR in Kerala and Andhra Pradesh Among the lowest in India

SRS SPECIAL BULLETIN ON MATERNAL MORTALITY, 2021–2023

INDIA'S NATIONAL MMR IS FALLING



Insights

REGIONAL GAP

- DISPARITY
- INEQUALITY

NEXT HEALTHCARE CHALLENGE

- DETECTION
- INTERVENTION

Sources: SRS Special Bulletin 2019–21 (RGI); UN MMEIG Report, April 2025; Million Death Study (2021 peer-reviewed analysis).

ROOT CAUSE

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Why DO Current Systems Fail?

India already has digital health infrastructure. The gap isn't digitization — it's prediction and field usability.

TODAY — ANMOL / REGISTERS

- HOME VISIT DATA ENTRY
- STATIC RECORD STORAGE
- NO PREDICTIVE SIGNAL

WHAT'S MISSING — SAHAI

- OFFLINE-FIRST FIELD CAPTURE
- COMPOSITE RISK SCORING (MVI)
- PRIORITISED, ACTIONABLE ALERT

Proof the model can work: Uttar Pradesh's ASHA Incentive Application disbursed ₹2,887 crore to 1.5 lakh ASHAs (Sep 2018–Mar 2022) — it digitized payments, but still computes no risk score.

THE SOLUTION

Meet SahAI

One offline-first app that turns a home visit into a predicted, prioritised action.



PROBLEM

A rising maternal risk factor appears during a home visit



AI ENGINE

MVI scores it across clinical, social, access & environment domains



ACTION

ASHA & PHC get a prioritised, offline-ready alert and referral path

END TO END FLOW

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The User Journey



PREGNANT WOMEN

Home visit & symptoms observed



ASHA WORKER

Captures data offline via SahAI



SAHAI

Computes MVI, flags risk, guides next step



DOCTOR / PHC

Receives referral with risk context



HEALTH ADMINISTRATOR

Tracks district-level outcomes on dashboard

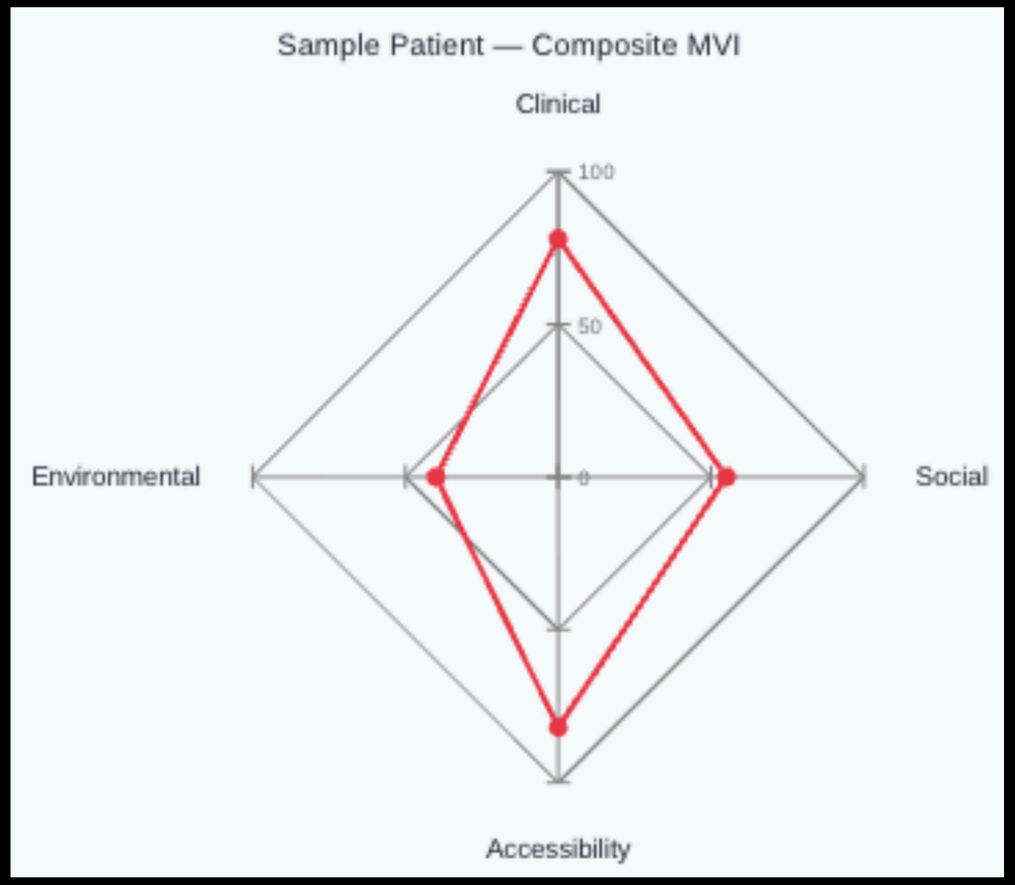
IMPROVED OUTCOME: A completed, on-time referral instead of a missed or delayed one — the exact gap the “three delays” model of maternal death identifies.

CORE IP

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The Maternal Vulnerability Index (MVI)

A composite score across four domains — seeded from published risk literature, recalibrated with pilot data



DOMAIN	EXAMPLE INPUTS
Clinical	BP trend, anaemia signs, prior C-section, gestational age at first ANC
Social	Age at pregnancy, birth order, education, household decision-making autonomy
Accessibility	Distance to CEmOC/blood-bank facility, transport, past referral non-completion
Environmental	Monsoon road-cut risk, seasonal migration status

AI ARCHITECTURE & OFFLINE SYNC

1. DATA COLLECTION

Voice + form capture on ASHA device, fully offline

2. PREPROCESSING

Cleans, structures, and encrypts field data locally

3. MODEL TRAINING

Risk models retrained on aggregated, anonymised pilot data

4. PREDICTION (MVI)

Composite score computed on-device or on sync

5. RECOMMENDATION ENGINE

Priority action + referral path suggested

6. DASHBOARD

PHC / district view once connectivity resumes

TECHNOLOGY STACK

FRONTEND

REACT NATIVE — LOW-END ANDROID SUPPORT

BACKEND

NODE.JS MICROSERVICES — MODULAR, SCALABLE

CLOUD

REGIONAL CLOUD (DATA-LOCALISED) — INDIA-HOSTED

DATABASE

SQLITE ON-DEVICE + POSTGRES SERVER-SIDE

ML MODELS

GRADIENT-BOOSTED MVI SCORER, LIGHTWEIGHT ENOUGH FOR EDGE INFERENCE

NLP

MULTILINGUAL ASR FOR VOICE-FIRST, LOW-LITERACY USE

OFFLINE ENGINE

LOCAL-FIRST STORE WITH CONFLICT-RESOLVED SYNC

SECURITY

END-TO-END ENCRYPTION, CONSENT-FIRST DATA MODEL

FEATURE SET

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OFFLINE MODE

FULL FUNCTIONALITY WITHOUT SIGNAL

VOICE ASSISTANT

HANDS-FREE, LOW-LITERACY FRIENDLY

RISK PREDICTION

MVI-DRIVEN EARLY WARNING

REFERRAL TRACKING

END-TO-END COMPLETION VISIBILITY

INCENTIVE AUTOMATION

TRANSPARENT, TIMELY DISBURSEMENT

DASHBOARD ANALYTICS

DISTRICT & STATE LEVEL VIEWS

MVI SCORING

COMPOSITE, MULTI-DOMAIN INDEX

LOCAL LANGUAGE SUPPORT

VOICE + TEXT IN REGIONAL LANGUAGES

LANDSCAPE

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Competitive Analysis

Solution	What it does	What it doesn't do	SahAI's edge
ANMOL / govt. apps	Records ANC visits, immunization into ABDM-linked records	No predictive risk scoring; connectivity-dependent	Offline-first + prediction layer on top
SMARThealth Pregnancy 2 (George Institute + Oxford)	AI-assisted screening for anaemia, hypertension, GDM; piloting in Haryana & Telangana	Single-condition clinical screening only; no incentive layer	MVI is multi-domain (social + access, not just clinical)
ARMMAN + Google DeepMind (mMitra, Kilkari)	ML-prioritised outreach calls at national scale	Engagement prediction, not individual clinical risk	Complementary — could integrate, not duplicate
SanRaksha (open-source)	Offline risk detection + dietary advice	Early-stage; no incentive automation or gov integration	Production-grade, incentive-integrated, ABDM-aligned
UP ASHA Incentive App	₹2,887 Cr disbursed to 1.5 lakh ASHAs (2018–22)	Payment tracking only — no clinical function	Extends this proven model + fuses risk prediction

SIZING

Market Opportunity

Methodology: seat-based B2G model tied to India's actual ASHA workforce, not a consumer app-store TAM.
UP already funds comparable digitisation (₹2,887 Cr via the ASHA Incentive App) through the same NHM PIP channel.



TOTAL ADDRESSABLE MARKET

10 lakh+ ASHA workers
All states/UTs under NHM (PIB, 2025)



STRATEGY FOR AT-RISK MOTHERS

EAG states priority
UP, Bihar, MP, Rajasthan, Odisha, Chhattisgarh



STRATEGY FOR ONBOARDING

2,000–5,000 ASHAs
1–3 pilot districts

Impact Metrics



REFERRAL COMPLETION RATE

Flagged high-risk case → reaches facility



TIME-TO-FLAG

Risk factor appears → ASHA/PHC alerted



DOCUMENTATION TIME SAVED

Hours/week freed per ASHA



DUPLICATE PAPERWORK REDUCED

Across ANMOL, registers, SahAI

A specific % reduction in maternal mortality. ARMMAN's own published work with Google DeepMind measured call-pickup improvements (up to 12% in specific time slots) as its outcome metric — not a mortality claim — even after a mature, multi-year deployment. We hold the same evidentiary standard from day one.

Buisness Model

Deliberately not leading with: a consumer subscription. ASHAs are unsalaried volunteers earning ₹2,000–₹15,000/month — charging them or their beneficiaries directly is both unrealistic and reputationally risky.

PRIMARY — B2G

STATE NHM PROGRAMME IMPLEMENTATION PLAN (PIP) BUDGET LINE — THE EXACT CHANNEL THAT FUNDED UP'S ASHA INCENTIVE APP.

SECONDARY — NGO / CSR

CO-FUNDS PILOT DISTRICTS BEFORE STATE PIP ADOPTION — MIRRORS HOW SMARTHEALTH PREGNANCY 2 AND ARMMAN FUNDED EARLY PILOTS.

TERTIARY — GRANTS

MULTILATERAL/GRANT FUNDING (GATES FOUNDATION, WHO, WORLD BANK) — REALISTIC ONLY POST-PILOT-VALIDATION.

Roadmap

PHASE 1

PROTOTYPE

MONTHS 0–3

PHASE 2

PILOT

MONTHS 4–9 — 1–2 BLOCKS

PHASE 3

DISTRICT DEPLOYMENT

MONTHS 10–18

PHASE 4

STATE DEPLOYMENT

YEAR 2–3

PHASE 5

NATIONAL SCALE

YEAR 3+

Named starting point: pilot districts drawn from EAG states (UP, Bihar, Odisha) where MMR is highest and the UP ASHA Incentive App precedent already demonstrates state-level willingness to fund similar digitisation.

Future Vision

- **DIGITAL TWIN FOR PREGNANCY**
- **FEDERATED LEARNING**
- **PREDICTIVE HEALTHCARE ECOSYSTEM**
- **AI HEALTHCARE NETWORK**
- **GLOBAL EXPANSION**

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Prevention Is Not a Feature. It's the Mission.

THANK YOU!

Every ASHA worker already carries the knowledge to save a life.
SahAI gives her the 30 seconds of warning she never had.

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